

Passivity-Based Decentralized and Centralized Control Using Predicted Control Barrier Functions

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Abstract

This report presents centralized and decentralized control strategies for multi-robot systems based on predicted Control Barrier Functions (CBFs). The centralized approach computes joint accelerations for all agents using a single quadratic program (QP), ensuring globally consistent safety and connectivity. The decentralized approach distributes computation across agents, each solving its own local QP based on neighborhood information. Both formulations employ predictive barrier functions to anticipate constraint violations and maintain smooth, collision-free, and connected motion.

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Chapter 1

Introduction

Multi-robot systems require coordination strategies that guarantee safety, connectivity, and convergence toward shared goals. Control Barrier Functions (CBFs) provide a powerful framework for enforcing safety constraints while allowing flexible motion control. This report compares centralized and decentralized implementations of predicted CBFs applied to double-integrator dynamics.

1.1 System Dynamics

Each robot is modeled as a double integrator:

$$\dot{x}_i = v_i, \quad \dot{v}_i = u_i \tag{1.1}$$

where $x_i, v_i, u_i \in \mathbb{R}^2$ denote the position, velocity, and acceleration of agent i , respectively.

Chapter 2

Centralized Predicted CBF Controller

In the centralized framework, a global controller solves one QP for all agents simultaneously.

2.1 Predicted Barrier Functions

Predicted barrier functions incorporate future motion via a prediction horizon T_{pred} .

Collision avoidance:

$$h_{\text{col}} = \|x_{ij} + T_{\text{pred}}v_{ij}\|^2 - d_{\text{min}}^2 \quad (2.1)$$

Connectivity maintenance:

$$h_{\text{conn}} = R_{\text{max}}^2 - \|x_{ij} + T_{\text{pred}}v_{ij}\|^2 \quad (2.2)$$

The CBF constraint ensures that h remains positive:

$$\dot{h}(x, v, u) + \alpha h(x, v) \geq 0 \quad (2.3)$$

2.2 Centralized Quadratic Program

At each time step, the controller solves:

$$\begin{aligned} \min_u \quad & \|u - u_{\text{nom}}\|^2 \\ \text{s.t.} \quad & \dot{h}_{\text{col}} + \alpha h_{\text{col}} \geq 0, \\ & \dot{h}_{\text{conn}} + \alpha h_{\text{conn}} \geq 0, \\ & \dot{h}_{\text{obs}} + \alpha h_{\text{obs}} \geq 0. \end{aligned} \quad (2.4)$$

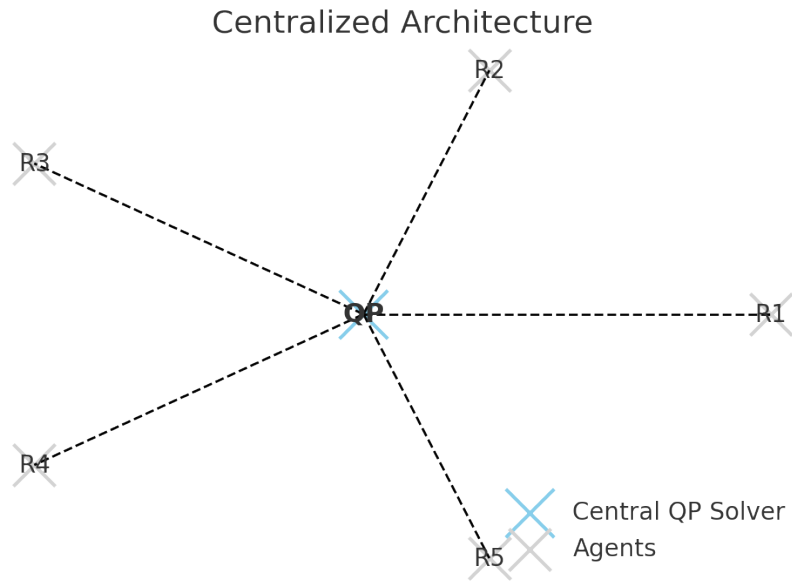


Figure 2.1: Centralized architecture: all agents' accelerations are optimized jointly.

2.3 Discussion

The centralized approach guarantees consistency and optimal cooperation, but computational complexity grows quadratically with the number of agents.

Chapter 3

Decentralized Predicted CBF Controller

In the decentralized strategy, each robot solves its own local QP using neighbor and obstacle information.

3.1 Local Formulation

Each agent i enforces local barrier functions for its neighbors $j \in \mathcal{N}_i$:

$$h_{\text{col}}^{ij} = \|x_{ij} + T_{\text{pred}}v_{ij}\|^2 - d_{\text{min}}^2 \quad (3.1)$$

$$h_{\text{conn}}^{ij} = R_{\text{max}}^2 - \|x_{ij} + T_{\text{pred}}v_{ij}\|^2 \quad (3.2)$$

The local CBF inequality remains:

$$\dot{h}^{ij}(x, v, u_i) + \alpha h^{ij}(x, v) \geq 0 \quad (3.3)$$

Each robot assumes neighbors' accelerations u_j remain approximately constant from the last step.

3.2 Local Quadratic Program

$$\begin{aligned} \min_{u_i} \quad & \|u_i - u_{\text{nom},i}\|^2 \\ \text{s.t.} \quad & \dot{h}_{\text{col}}^{ij} + \alpha h_{\text{col}}^{ij} \geq 0, \quad \forall j \in \mathcal{N}_i \\ & \dot{h}_{\text{conn}}^{ij} + \alpha h_{\text{conn}}^{ij} \geq 0 \end{aligned} \quad (3.4)$$

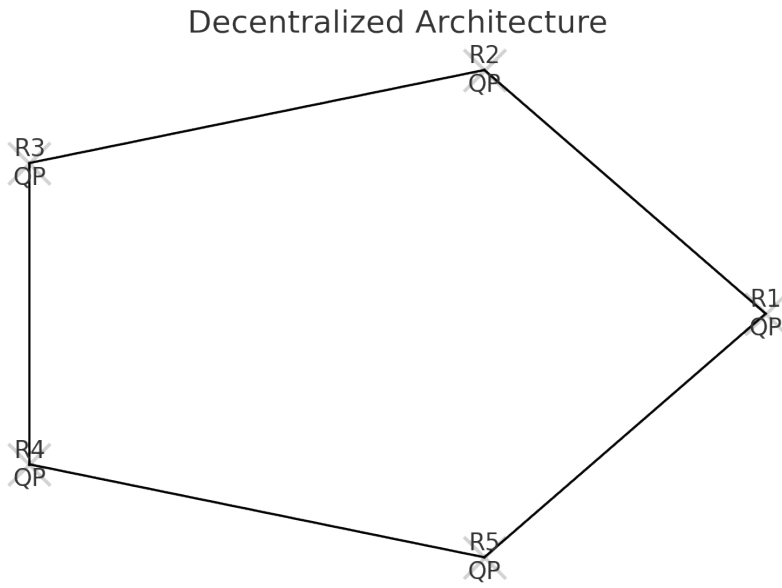


Figure 3.1: Decentralized architecture: each agent solves its own local QP using neighbor data.

3.3 Discussion

The decentralized approach drastically reduces computational cost and scales well with team size, though perfect global consistency cannot be guaranteed.

Chapter 4

Comparison of Strategies

Aspect	Centralized	Decentralized
Decision variable	All agents' accelerations	Local u_i per agent
Constraint coupling	Global (all pairs)	Local (neighbors only)
Computation	$O(N^2)$	$O(N)$
Slack variables	Optional (soft constraints)	None (hard constraints)
Communication	Full network	Local neighborhoods
Scalability	Poor for large N	Excellent
Safety	Globally coordinated	Locally predictive

Table 4.1: Comparison between centralized and decentralized predicted-CBF control.

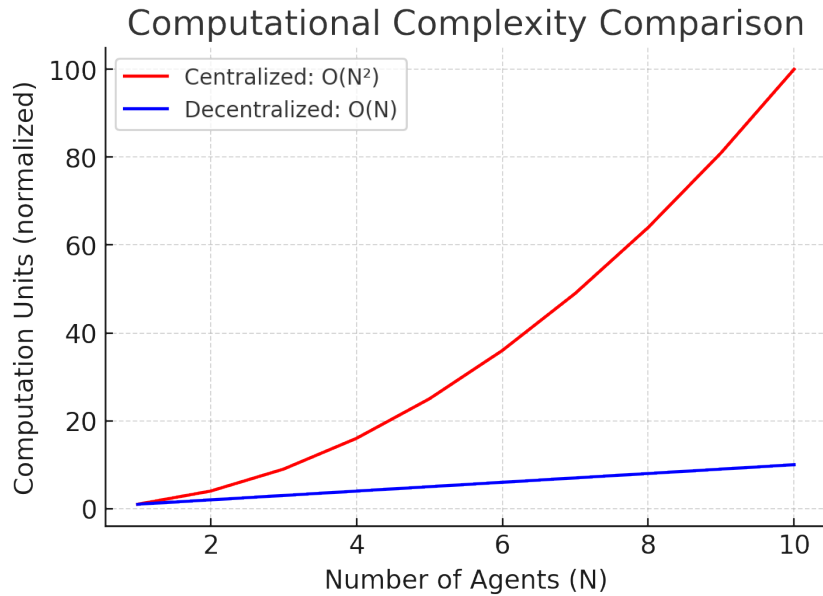


Figure 4.1: Comparison of computational and communication scalability.

Chapter 5

Possible Improvements and Extensions

- Incorporate formation or cohesion terms in the nominal control law.
- Use distributed estimators for λ_2 to adapt connectivity constraints dynamically.
- Combine Control Barrier Functions with Control Lyapunov Functions (CBF-CLF) for stability guarantees.
- Employ consensus-based filters to estimate neighbors' accelerations.

Chapter 6

Conclusion

Both centralized and decentralized predicted-CBF controllers ensure safety and connectivity in multi-robot systems. The centralized version offers global coordination but lacks scalability, while the decentralized method achieves efficient, distributed safety enforcement suitable for large teams.